

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here’s a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])
by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;
Thu, 10 Aug 2023 07:00:16 -0700 (PDT)
Received: by mail.example.com with SMTP id almr1243772ywh.51;
Thu, 10 Aug 2023 07:00:15 -0700 (PDT)
Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Message ID	<010901a031da7783-d1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com>
Created at:	Mon, Aug 24, 2026 at 9:09 AM (Delivered after 0 seconds)
From:	HyugaLife <support@hyugalife.com>
To:	manideepchinthalapalli2007@gmail.com
Subject:	Your order #1002951434 is out for delivery today 📦
SPF:	PASS with IP 76.223.180.121 Learn more
DKIM:	'PASS' with domain hyugalife.com Learn more
DMARC:	'PASS' Learn more

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Delivered-To: manideepchinthalapalli2007@gmail.com
Received: by 2002:a05:7301:1821:b0:327:e43d:e952 with SMTP id md33csp2787773dyb; Sun, 23 Aug 2026 20:39:57 -0700 (PDT)
X-Received: by 2002:a17:90b:4a0d:b0:38e:5b59:c2ff with SMTP id 98e67ed59eid1-395c3376963mr37637345a91.3.1787542797001; Sun, 23 Aug 2026 20:39:57 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787542796; cv=none; d=google.com; s=arc-20260327; b=Tc1Rn9MPYK23C5fIN9rzfId8YeDjJceg8DTvCk6YhBbVH1zooFckohndP0q3MFQTHd xPFGJoztewJBo9PxCaZFA73Fg7eq7oHwX1S+Y+I12LZohWZdteapRLZFs72+49Jj2Er2 StD2zvVYJ39P8ajGHKWnw5ig+WAGdMCTg6hte/xrZf2YYpakCga41f46X6iCHwmlly4 NGotbRatB84tvZNJJkrYmGK10RoFyxverMMDrX9/ZrNuLVDfB+3ktUY4cV09c8sQK1Qx TBxvkbF1Km5ADMXJnrXz/1aFb27vAautwChd/2R+MjYidwt0x1aTCf53The22n1uPs 9a+w==



TOOLBOX[®]

SUPERTOOL

Login

Email Header Analyzer

Paste Header:

```
</p>
</td>
</tr>
</table>
</td>
</tr>
</table>
</div>
</body>
</html>
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

Your IP is: 223.187.117.50 | Contact Terms & Conditions Security API Privacy Phone: (866)-696-9652 | © Copyright 2004-2025 MXToolbox, Inc. All rights reserved. US Patents 10039363 B2 & 11461730 B2



TOOLBOX[®]

SUPERTOOL

Login

Header Analyzed

Email Subject: Your order #1002951434 is out for delivery today 📦

Analyze New Header

Delivery Information

DMARC Compliant

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received

Delay:

1 seconds

From: c180-121.smtp-out.ap-south-1.amazonaws.com to mx.google.com

to:

Relay (Seconds)

0 0.5 1 1.5 2 2.5

0 0.5 1 1.5 2 2.5

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	c180-121.smtp-out.ap-south-1.amazonaws.com 76.223.180.121	mx.google.com	ESMTPS	8/24/2026 3:39:56 AM	✓
2	1 Second		2002:a05:7301:1821:b0:327:e43d:e952	SMTP	8/24/2026 3:39:57 AM	

SPF and DKIM Information

dmarc:hyugalife.com

Hide

Solve Email Delivery Problems

v=DMARC1; p=none;

Tag	TagValue	Name	Description
Tagv	Tag Value	NameVersion	DescriptionIdentifies the record retrieved as a DMARC record. It must be the first tag in the list.
Tagp	Tag Value	NamePolicy	DescriptionPolicy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.

Test	Result
Status NameDMARC Policy Not Enabled	ResponseDMARC Quarantine/Reject policy not enabled
Status NameDMARC Record Published	ResponseDMARC Record found
Status NameDMARC Syntax Check	ResponseThe record is valid
Status NameDMARC Multiple Records	ResponseMultiple DMARC records corrected to a single record.
Status NameDMARC External Validation	ResponseAll external domains in your DMARC record are giving permission to send them DMARC reports.

Your DNS hosting provider is "Amazon Route 53" [Need Bulk Dns Provider Data?](#)

Reported by ns-193.awsdns-24.com on 8/27/2026 at 3:47:45 AM (UTC 0). [just for you](#) [Transcript](#)

spf:ap-south-1.amazonses.com:76.223.180.121

Show

v=spf1 ip4:54.240.100.0/23 ip4:76.223.152.0/23 ip4:76.223.176.0/24 ip4:76.223.180.0/23 ip4:76.223.188.0/24

dkim:hyugalife.com:t6epf6eect2a3ey3qirxmhofhs6htfgy

Show

Dkim Public Record:

p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtdJ5Y45aAwdxBmb3gNFNw+Nxx5YrYr4oZCeI9aq8BR+xt1PUjJLYQyF2vdFFI

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=t6epf6eect2a3ey3qirxmhofhs6htfgy; d=hyugalife.com; t=178;

dkim:hyugalife.com:t6epf6eect2a3ey3qirxmhofhs6htfgy

Show

Dkim Public Record:

p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtdJ5Y45aAwdxBmb3gNFNw+Nxx5YrYr4oZCeI9aq8BR+xt1PUjJLYQyF2vdFFI

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=t6epf6eect2a3ey3qirxmhofhs6htfgy; d=hyugalife.com; t=178;

dkim:amazonses.com:rlntogby6xsxlfvnyxwnvhttakdsqto

Show

Dkim Public Record:

p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDk6/ZugK+X3pK5nwBw0tQMxf/vi/AjQ5UY5kVJYmiG26V19ahULB6KbU/mFc dg3hu

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=rlntogby6xsxlfvnyxwnvhttakdsqto; d=amazonses.com; t=178;

Headers Found

Header Name	Header Value
Delivered-To	manideepchinthalapalli2007@gmail.com
X-Received	by 2002:a77:90b:4a0:d:b0:38e:5b59:c2ff with SMTP id 98e67ed59e1d1-395c3376963mr37637345a91.3.1787542797001; Sun, 23 Aug 2026 20:39:57 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=1787542796; cv=none; d=google.com; s=arc-20260327; b=TdRn9MPYK23C5fN9rzfid8YeDjJceg8DTvCk6YhBbYHHzoFokohndP0q3MFQTHd xPFGJozteWJB09PxCazFA73Fgq7wq70HxK1S+Y+iiZLZoWZdteapRLZF7s72+49JZEr2 StD2zvVYJ39P8aGhKWNw5ig+wAGdMCTg6hte/xzZi2YyPaKcga4f46X6iCHWwmLjy4 NG0tbRAiEB84tvZNNJkRYmGK10RofYxeverMMdrX9/ZrNuLVDFB+3ktUY4cV09c8sQKIQx l8xsykbEjKm5ADMXJnrXz/igfb2ZyAqutWChd/28+MJYjdWt0xpATCf53THq22pl uPs 9a+w==
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:message-id:content-transfer-encoding:mime-version:subject:to:from:date:dkim-signature:dkim-signature; bh=cC8GxsbotAn7pohlaK0G04rWvhq3PTXCvm6t7HmqzQY=; fh=OXO/8OkGc7QctTccD/YR8Bv/RFW6ePslglcxT5ODYyA=; b=jDUPRYyvhDCTUr6D02p7J5mHMENZsy4XycudlWvthjvVXCbm7FcDzOvIno3FrQY9l hABIGy/+dDzMARx+dg1sk5falc8zW5Pez6TFA/t5Ewt/tiLxxkxAlln37oyKUJkb6MYK sdxHoGrqd8ljkJO hA3sldOeCFpQXmDoGU6TbC5DwztJTpHYb+KkHdN8fcBFC42ktaft6 748WQNCeH3+K74tbBx5txvET5yVAtjopjEHEh7GQyLCAedtf8vXhyB8T5bw5Fv7wRMXZM 7OJZ/amU+J+c6+lvPwTY824lBdSSSAEgHdTtc59VseNLRIMXzYw/wPhF0smSlsOB9je ofOg==; data=google.com
ARC-Authentication-Results	i=1, mx.google.com; dkim=pass header.i=@hyugalife.com header.s=t6epf6eect2a3ey3qirxmhoHfs6htfgy header.b=aqHUXCTQ; dkim=pass header.i=@amazonses.com header.s=rIntogby6xsxfnvyxwnvhttakdsqto header.b="2D/p6RnN"; spf=pass (google.com: domain of 010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com designates 76.223.180.121 as permitted sender) smtp.mailfrom=010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com; dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=hyugalife.com
Return-Path	<010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com>
Received-SPF	pass (google.com: domain of 010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com designates 76.223.180.121 as permitted sender) client-ip=76.223.180.121;
Authentication-Results	mx.google.com; dkim=pass header.i=@hyugalife.com header.s=t6epf6eect2a3ey3qirxmhoHfs6htfgy header.b=aqHUXCTQ; dkim=pass header.i=@amazonses.com header.s=rIntogby6xsxfnvyxwnvhttakdsqto header.b="2D/p6RnN"; spf=pass (google.com: domain of 010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com designates 76.223.180.121 as permitted sender) smtp.mailfrom=010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com; dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=hyugalife.com
DKIM-Signature	v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=t6epf6eect2a3ey3qirxmhoHfs6htfgy; d=hyugalife.com; t=1787542796; h=Date:From:To:Subject:MIME-Version:Content-Type:Content-Transfer-Encoding:Message-ID; bh=WdXEqlLpikxvHd/4x/fJBXivmSiYWAYDcAndJ9JGQ=; b=aqHUXCTQCs8H87nTDZyTkyQdLJTrz/7WLCF/9Yj9EQGFidat4rK7Q/O5iOeWM 0etpmNo61AqV8+PZUUVWCvH1KN7Mo0DssjVCSebQx/QuNhODV+fWY4nadpvQlvdGeT ZbgcR29p/q38Jn5QLCDnDxO24xANYUJSSxa2nh2 Z6pF0vq9U5k3P0kDAooB1+GmWu7 YC3kjY6lrDPQwgOzMZuWY7S5z/WJ/Glc/HEK3597/FDixXAAtv79hFNqBDpxU8dcGC nD8Skdtfk00Nc/hyYwTFVmGITDrKNI+4jiXblydtOlGJlcXN80hGr6gyZsPrDP65sQ n2WqaKeH0g==
Date	Mon, 24 Aug 2026 03:39:56 +0000
From	HyugaLife <support@hyugalife.com>
To	manideepchinthalapalli2007@gmail.com
Subject	Your order #1002951434 is out for delivery today 📦
MIME-Version	1.0
Content-Type	text/html; charset=UTF-8
Content-Transfer-Encoding	quoted-printable
Message-ID	<010901a031da7783-df1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com>
Feedback-ID	:1.ap-south-1.XGasS3NW/HJSh0rx7XMKz1D8eoCV9CADSytcH/UUM=AmazonSES
X-SES-Outgoing	2026.08.24.76.223.180.121

X-SES-Outgoing	2026.08.24-76.223.180.121
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Received Header

```

Delivered-To: manideepchinthalapalli2007@gmail.com
Received: by 2002:a05:7301:1821:b0:327:e43d:e952 with SMTP id md33csp2787773dyb; Sun, 23 Aug 2026 20:39:57 -0700 (P
X-Received: by 2002:a17:90b:4a0d:b0:38e:5b59:c2ff with SMTP id 98e67ed59e1d1-395c3376963mr37637345a91.3.17875427970
ARC-Seal: i=1; a=rsa-sha256; t=1787542796; cv=none; d=google.com; s=arc-20260327; b=TcIRn9MPYK23C5fIN9rzfId8YeDjJce
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:message-id
ARC-Authentication-Results: i=1; mx.google.com; dkim=pass header.i=@hyugalife.com header.s=t6epf6eect2a3ey3qirxmhof
Return-Path: <010901a031da7783-d1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com>
Received: from c180-121.smtp-out.ap-south-1.amazonaws.com (c180-121.smtp-out.ap-south-1.amazonaws.com. [76.223.180.
Received-SPF: pass (google.com: domain of 010901a031da7783-d1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.a
Authentication-Results: mx.google.com; dkim=pass header.i=@hyugalife.com header.s=t6epf6eect2a3ey3qirxmhofhs6htfgy
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=t6epf6eect2a3ey3qirxmhofhs6htfgy; d=hyugalife.com
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=r1ntogby6xsxlfvnyxwvvhhtakdsqto; d=amazonses.com
Date: Mon, 24 Aug 2026 03:39:56 +0000
From: HyugaLife <support@hyugalife.com>
To: manideepchinthalapalli2007@gmail.com
Subject: =?UTF-8?Q?Your_order_#1002951434_is_out_for_delivery_today_=F0=9F=9A=9A?=
MIME-Version: 1.0
Content-Type: text/html; charset=UTF-8
Content-Transfer-Encoding: quoted-printable
Message-ID: <010901a031da7783-d1388985-1dae-4a13-be58-bdaaac7da51b-000000@ap-south-1.amazonaws.com>
Feedback-ID: ::1.ap-south-1.XGas53NW/HJSh0rx7rXMKz1D8eoCV9CAD5ytch/UUM=:AmazonSES
X-SES-Outgoing: 2026.08.24-76.223.180.121

<!DOCTYPE html>
<html
  lang=3D"en"
  xmlns=3D"http://www.w3.org/1999/xhtml"
  xmlns:o=3D"urn:schemas-microsoft-com:office:office"
>
<head>

```